

**Small Object Detection for Detecting Floating Bottles in Inland
Water Bodies using Attention-Based Improved YOLOv5**

A PROJECT REPORT

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BONAFIDE CERTIFICATE

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ABSTRACT

Water pollution has emerged as a significant environmental threat that requires attention at all levels, from inland water bodies to oceans. While numerous high-value marine equipment have been developed and deployed, it remains critical that flowing waters and inland water bodies do not accumulate and carry additional waste. Object detection techniques are essential for such initiatives, and developing them for real-time deployment poses a challenge. In this study, we focus on developing a model for detecting floating bottles, with a specific emphasis on small object detection. Our study employs the FloW-IMG dataset, provided by ORCA-Uboat. To improve the accuracy of the model, we utilize a transfer learning approach, where the base model YOLOv5 is pre-trained on COCO. Additionally, we enhance the model architecture by including two supplementary modules: a transformer-based C3TR and CBAM attention mechanism to enrich feature extraction. Our experimental results demonstrate that the proposed model achieves a best mean average precision (mAP) score of 45.6%, surpassing the previously attained highest accuracy on the dataset by 5%, in addition to a high recall and precision.

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CHAPTER - 1

INTRODUCTION

In today's world, we are inundated with distressing reports concerning the devastating impact of water pollution on the ocean vitality. This insidious threat has reached epidemic proportions and demands immediate action to be taken. The dire consequences of neglecting this issue are simply too grave to ignore. This increasing pollution alarmingly threatens the marine ecosystems and beings, disrupting the natural order [1]. According to a study conducted in 2018, plastic waste that is left unnoticed is rapidly accumulating in ocean with the biggest garbage patch called The Great Pacific Garbage Patch that is situated at Pacific Ocean [2]. There is a forecasted accumulation of more than 129,000 tonnes of garbage in The Great Pacific Garbage Patch, representing a staggering quantity [2]. Although there is a significant amount of effort being put on cleaning the floating marine waste, it is equally necessary to prevent the waste from reaching the oceans. In-land generated waste that reach the oceanic waters contribute consequential amount to the marine floating debris hence efforts driven towards clean flowing-water streams and lakes are equally crucial [3].

In recent years, significant strides have been made in addressing the pervasive issue of plastic pollution, with innovative methods being developed to combat this environmental threat. Utilization of cutting-edge technology such as robots, Unmanned Surface Vehicles (USVs), and Artificial Intelligence to collect floating garbage from water bodies are amongst the recent efforts. This represents a promising solution in the ongoing fight against water pollution, and has the potential to make a substantial impact in preserving our planet's delicate ecosystem. IWSCR [4] is a smart and automatically operating USV that was developed to clean water surface waste. ORCA-Uboat [3], a China based venture that supplies USVs to clean floating plastic waste from water bodies, has products have been deployed in hundreds of bodies of water in ten countries. Advancements in Computer Vision, a field of

Artificial Intelligence that empowers machines to obtain information from visual inputs, is playing a vital role in the development of robust USVs to make water sanity management thoroughly viable. Object detection, a sub-field of computer vision, enables machines to recognize patterns and locate specific objects within an image or video feed. With deep learning techniques and access to large amounts of labeled data, USVs equipped with computer vision and object detection capabilities can collect floating plastic garbage from water bodies, helping to mitigate the impact of plastic pollution on the environment. Such systems might be able to infer information way faster than humans, but it is still challenging to detect various objects from visual media in real-time. The major challenge lies in the area of coverage. In a typical scene with plastic waste floating in water bodies, the waste generally consists of only a few pixels in an image which is very challenging to capture and interpret with normal object detection. To deal with this niched-down detection problem, small object detection techniques come into picture [5].

Small object detection is a sub field of Computer Vision where the systems are designed to be able to detect relatively diminished objects in a visual content. ‘Small objects’ are the objects which span, relatively, a smaller fraction of the picture area, typically within 32×32 pixels [3]. Small object detection is a hot topic in current time with an array of real-world applications in the fields of defense, traffic management, rescue, any many more [6], [7]. However, one of the major challenges to the small object detection problem is that we usually do not have sufficient examples of the small object in the training examples. Due to small objects being so insignificant in size compared to the rest of the image itself, feature maps generated by basic CNNs don’t encode sufficient information about them [8]. Also, the context information for small objects is mostly lacking in most of the datasets. Refining small object detection methods for complex scenes like water surfaces is all the trickier.

Keeping all the above-mentioned problems in mind, in this work, in our work we focus on the problem of detecting floating objects on water surface using small object

detection methods, where the primary focus will be to achieve good performance in realistic frame rates.

The major contributions of this work include-

1. The best mAP score achieved by the proposed model is 45.6%. Transfer learning-based YOLOv5 model is adapted and fine-tuned for floating bottle detection.
2. The YOLOv5 model is enhanced by use of C3TR and CBAM modules in addition to use of Hardswish activation. This contributes increase in accuracy of 0.06% in comparison to actual YOLOv5 model.
3. Proposed model surpasses the previously attained highest accuracy on the FloW-IMG dataset by 1.8%.
4. Three activation functions, namely Swish, Leaky ReLU and Hardswish are compared to enhance model performance.

The outline of the project is as follows: Chapter 2 reviews the related work in floating waste detection in water bodies and presents a study over different small detection techniques; Chapter 3 mentions the software specifications required for undertaking this project; Chapter 4 explains the proposed system architecture in this work; Chapter 5 covers the specifics of the project implementation; Chapter 6 discusses the various test metrics for the system; The obtained results are summarized in Chapter 7; Finally, Chapter 8 presents our conclusion and future scope.

CHAPTER - 2

LITERATURE REVIEW

Anthropomorphic activities have constantly been the primary cause of many environment hazards including pollution of water bodies. In spite of the continuous efforts of many scientists, philanthropists and even volunteers, with billions of dollars invested in the cleaning efforts, this water pollution lives on and continues to increasingly harm the aqua-life. We identify one of the best resorts to tackling this problem as described in [3], is to deploy multiple cleaning units along the course of flow of water, and in inland water bodies like lakes, where water is continuously monitored and cleaned, hence preventing the debris from reaching the oceans. In this study, we assume the water waste to be small objects in comparison to the environment, as approached by [9]. A compact literature study has been presented in the following sections giving an overview of the existing research on the detection of water surface waste using computer vision and deep learning techniques, and how these techniques have evolved over time. The study is divided as follows – *Section 2.1.* describes the ongoing-research in the area of water waste handling systems; and *Section 2.2.* covers the advancements in different Small Object Detection techniques.

2.1. Floating Water Waste Detection

One of the early works published in 2019, proposed an improved version of the Faster R-CNN algorithm for real-time water surface object detection by use of explicit Region Proposal Network [10]. In 2020, [11], [12] proposed novel approaches for detecting trash on water channels using attention neural networks and deep transfer learning. AquaVision [12], an automated waste detection system that employs deep transfer learning and transferable feature maps. In 2021, several works like [13], [14] were published on the topic of automated detection of floating debris in water bodies. YOLOv3 is used as target identification and classification model in both. A network fusion approach for riverine floating debris monitoring is presented

in [15] that is developed on top of YOLOv4 model. FMA-YOLOv5s [16], an improved YOLOv5s algorithm for floating debris in waterways includes an additional feature map attention (FMA) layer at the end of the backbone, and achieves 79% mAP. Another paper published in 2021 introduced FloW [3], a dataset and benchmark for floating bottles detection in inland waters. Two papers published in 2021 proposed methods for detecting floating marine litter using optical remote sensing data. A review of floating marine litter detection algorithms and techniques is provided by [17], while [18] proposed a method for detecting macro floating objects on coastal water bodies using Sentinel-2 data. Additionally, [19] proposed a robust small object detection method on water surfaces through the fusion of camera and millimeter-wave radar. In 2022, [20] and [21] propose automated solid waste detection systems for riverine management. In [20] many improvements in the YOLOv4 architecture have been made to achieve an accuracy of 89% mAP. An image segmentation-based approach is presented in [21] for river debris detection.

While recent advancements in floating debris detection and removal from water bodies have been highlighted in the existing literature, there are still several gaps and opportunities for future research. One of the primary challenges is the lack of standardization in the datasets used by various approaches, which makes it difficult to compare the results of different methods. This limitation hinders the potential for wider adoption and use of these methods. Furthermore, most of the proposed approaches are based on deep learning techniques that require substantial data for training and testing. However, the availability of standard data for such models is limited, which presents a significant challenge. In addition, some methods rely on expensive sensing equipment that may not be feasible for practical applications. Despite these challenges, the literature review showed that the YOLO-based models are the most practically adopted for floating debris detection in water bodies. Nevertheless, there is still significant scope for research to improve the accuracy of these methods and make them more accessible to practical applications.

2.2. *Small Object Detection Techniques*

Small object detection is a subset of ever-famous object detection problem in Computer Vision, that has gained a lot of importance in the recent times. As a complex and challenging problem, small object detection forms the basis for many Computer Vision tasks like instance segmentation, image captioning, object tracking [22]. Small objects are mostly defined as one of the follows- objects relatively smaller than other objects in the same image or any objects that occupy less than 32*32-pixel area [22],[23].

Perceptual Generative Adversarial Networks (PGAN) [24] for small object detection was proposed in 2017. PGAN introduces a new loss function to train the generator to generate perceptually realistic images that can help detect small objects. In the same year, Mask R-CNN [25] was introduced, which extended the Faster R-CNN architecture by adding a mask branch for pixel-level object segmentation. These two methods showed promising results in small object detection and have since then been used as baseline models in subsequent research. Multiscale feature extraction is essential for small object detection, as it allows the network to capture objects at different scales. In 2018, a Small Object Detection with Multiscale Features (SOD-MF) [26] method was proposed that used a multiscale feature fusion module to extract features from different scales. In 2019, an improved Faster R-CNN method was proposed, which used a feature pyramid network (FPN) and a novel receptive field enhancement module to enhance small object detection [27]. In 2020, several studies evaluated deep learning methods for small object detection [22], [28], and recent advances were reviewed. Attention-based methods were found to perform well in small object detection tasks [29]. Furthermore, a study of deep learning methods for small object detection, conducted in 2021, showed that YOLOv5 and Mask R-CNN were among the best-performing methods [8]. In recent years, there has been an increasing interest in developing lightweight models for small object detection. In

2021, several approaches were proposed, such as [29], [30], [31], based on attention feature fusion. In the same year, an improved Mask R-CNN [32] was proposed for aircraft detection in remote sensing images which makes use of instance segmentation and is robust to dense targets and complex background. In 2022, several lightweight small object detection methods were proposed, such as LDS-YOLO [33] and SF-YOLOv5 [34] for improved feature fusion. Works [35] and [36] purely emphasize the effectiveness of data augmentation techniques for small object detection in computer vision. Standard augmentation methods are combined with YOLOv5 model in [35], while in [36] a GAN-based pipeline specifically designed for small object detection is adapted, which included synthetic image generation and random augmentation techniques. The tailored pipeline may provide greater benefits due to its ability to increase the diversity and quantity of training data. In 2023 attention-based approaches remain popular as in [37], in which the authors use a transformer-based module and convolution block attention module to enhance the feature map selection process showing promising results on detection of small and medium-sized target objects.

Based on the current state of research in small object detection, there are several deep learning methods that have shown promising results. However, YOLOv5 has been successfully combined with standard data augmentation techniques, which may make it a more practical option for detecting floating bottles on inland water bodies from images in a realistic amount of time. By exploring the use of YOLOv5 and data augmentation techniques for detecting floating bottles, it may be possible to achieve good detection performance while also keeping computational requirements manageable.

In conclusion, this project aimed to develop a system for detecting floating bottles in inland water bodies using small object detection techniques. The literature study explored similar systems and various detection techniques, leading to the selection of

the FloW-IMG dataset [3] and YOLOv5 as the base model for this project. However, the proposed baseline models have a relatively low accuracy score that is not practically suitable for an application such as ours. To improve the accuracy of the model, we explore inclusion of Transformer-based modules, along with data augmentation techniques. The implementation of these methods will be evaluated and compared with the baseline performance of YOLOv5. We believe that the proposed enhancements can lead to a significant improvement in the detection accuracy of the system, which can have important applications in environmental monitoring and pollution control.

CHAPTER – 3 SYSTEM SPECIFICATIONS

3.1 Software requirements

The present undertaking is a software-based project that does not necessitate any hardware components. The implementation phase was carried out using the Python programming language, within the Google Colaboratory environment. The programs were executed using a GPU setting, utilizing the hardware accelerator feature. The Colab platform provides access to an Nvidia Tesla K80 GPU. The project was conducted on a local system with 16 GB of RAM.

As the system's implementation is entirely software-based, it is platform-agnostic, capable of being deployed on online Integrated Development Environments (IDEs) like Google Colab, as well as on local IDEs such as Jupyter Notebook and PyCharm.

CHAPTER - 4

SYSTEM DESIGN

The system design section provides an in-depth explanation of the proposed system. In Section 4.1, a detailed description of the dataset used is presented, along with sample images that highlight the challenges of working with small objects. Moving on to Section 4.2, the top-level workflow of the proposed design is outlined, including the key steps involved in implementing the system. Subsequently, Section 4.3 delves into the architecture and specifications of the deep learning model that has been tailored to meet the requirements of the project.

4.1. Dataset

The dataset used in our work is FloW-IMG, that is provided by ORCA-Uboat [3]. It consists of 2,000 .jpg photographs containing over 5,000 labeled floating waste (small objects) and annotations are provided to locate objects in the given image with proper co-ordinates and bounding box parameters. Also, 200 videos that are captured real-time are provided without annotations. The dataset consists of a single class of floating water waste which is plastic bottles, and the images are primarily captured in inland water bodies like lakes from a camera attached to an Unmanned Surface Vehicle. The images have a resolution of 1280*720 (mostly) and 1280*640. The small objects considered are defined such that they span less than 32*32-pixel area. Sample images from the FloW-IMG dataset are given in *Figure 1*. Different scenarios and challenges in the course of performing small object detection are depicted in the same.



1.a



1.e



1.b



1.f



1.c



1.g



1.d



1.h



1.i



1.k



1.j



1.l

Figure 1: Sample images from FloW-IMG dataset

An illustrative dataset showcasing various challenges encountered during object detection in water bodies is presented in figure 1.a. where a bottle is observed floating on the water surface. However, identifying and detecting such objects is not always straightforward, as illustrated in the subsequent figures. In figure 1.b, the highlighted region demonstrates an example of a small object that is difficult to interpret due to its diminutive size. Similarly, in figure 1.c, a bottle is shown camouflaging with the background, making it hard to differentiate. Figure 1.d highlights the effect of water reflections, which create distracting contrasting patterns and make it challenging to locate objects, particularly when they are camouflaged along the water boundary. Additionally, the images in Figure 1.e are blurred and shaky due to the movement of the unmanned surface vehicle (USV) on the water surface, which can lead to difficulties in object detection. The complex mosaic

patterns that arise due to light reflections and wave motion of water molecules are illustrated in figure 1.f. The mosaic patterns can diminish the clarity of the object's boundaries and shape, making it difficult to detect objects of interest. Figure 1.g shows an image with clustered bottles in a small patch, and Figure 1.h highlights how plastic waste can appear very similar to or aligned with the reflections of other objects, making detection a challenging task. Moreover, figure 1.i illustrates how the mid-day sun, directly above the water body, can create white patches in most areas of the water surface, taking away the color and clarity of floating objects. Images with high contrast, such as the one shown in figure 1.j, can be helpful in enhancing small object boundaries. However, in cases where images are cluttered with different textures, increased contrast can diversify the focus. Figure 1.k shows an example where small objects occur at varying distances and therefore measure up to different sizes. Hence, the detection mechanism should be robust to the depth information of different objects in view. Finally, figure 1.l demonstrates an example where the object of interest is extremely illuminated, making its color and on-surface details difficult to discern. Detection of such objects is only possible if an idea of the general shape of the object is known.

4.2. High-level System Design

The high-level architecture, illustrated in Figure 2, comprises various modules ranging from data collection and pre-processing to detection. The initial step involves collecting data from the ORCAUBOAT official website, which provides images and PASCAL VOC XML annotations. Subsequently, the annotations data undergoes parsing through a module to obtain the required YOLO standard format for use in YOLO. Further, the images are warped to fit the YOLOv5 input dimension of 640 x 640 pixels. The training data, along with corresponding YOLO annotations, is then fed to the YOLOv5 model, whose parameters are discussed in the implementation section of this document. For testing, the YOLO-formatted test set from the dataset is

used. Visualization tools such as ClearML and Comet are utilized to monitor the machine learning program's progress through the iterations.

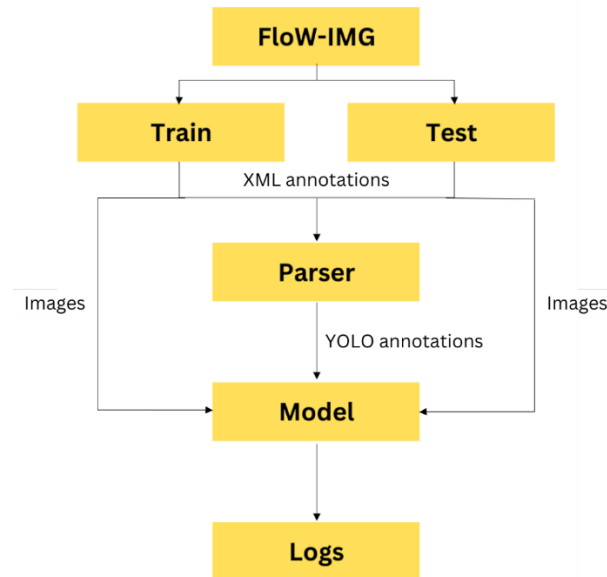


Figure 2: Flow chart showing the high-level system design

4.3. Model Architecture

The task of object detection, as tackled by two-stage object detection techniques, involves two sub-problems: the detection of potential object regions and the classification of the image within those regions into object classes. Two-step OD algorithms employ a Region Proposal Network that suggests regions of interest which are then fed to a classifier. This technique is time-consuming as it necessitates multiple iterations for each image. In contrast, YOLO [38], a single-stage detection technique that originated around 2015, utilizes a single end-to-end neural network to perform both of the aforementioned tasks, making it significantly faster.

The YOLO model functions by partitioning the input image into 'n' $m \times m$ (square) grids. Each grid is tasked with identifying and localizing the objects within its boundaries. The corresponding grid then predicts bounding box coordinates, object labels, and the likelihood of the object's presence in the grid. While this approach is less computationally demanding, it results in many duplicate predictions as multiple grids can present the same object with different bounding box predictions. YOLO resolves this issue by utilizing Non-Maximal Suppression, which suppresses or eliminates bounding boxes with lower object probability values. This process is repeated multiple times until the final bounding boxes for each identified object are determined. The model produces three outputs: the bounding box, its class label, and its probability score.

The YOLO architecture can be divided into three primary components: a backbone neural network, the model neck, and the model head. The backbone is usually a pre-trained neural network that extracts feature maps from the input image. Feature maps decrease the image's spatial resolution and focus only on its essential features. The model neck forms feature pyramids that function as feature extractors at various levels for a single-scale input image. Feature pyramids play a crucial role in enabling the model to generalize well to detecting objects of varying sizes within the image. The model head applies anchor boxes to feature maps and generates the final output. Anchor boxes are predetermined bounding boxes of a certain height and width selected based on the object sizes in the training data.

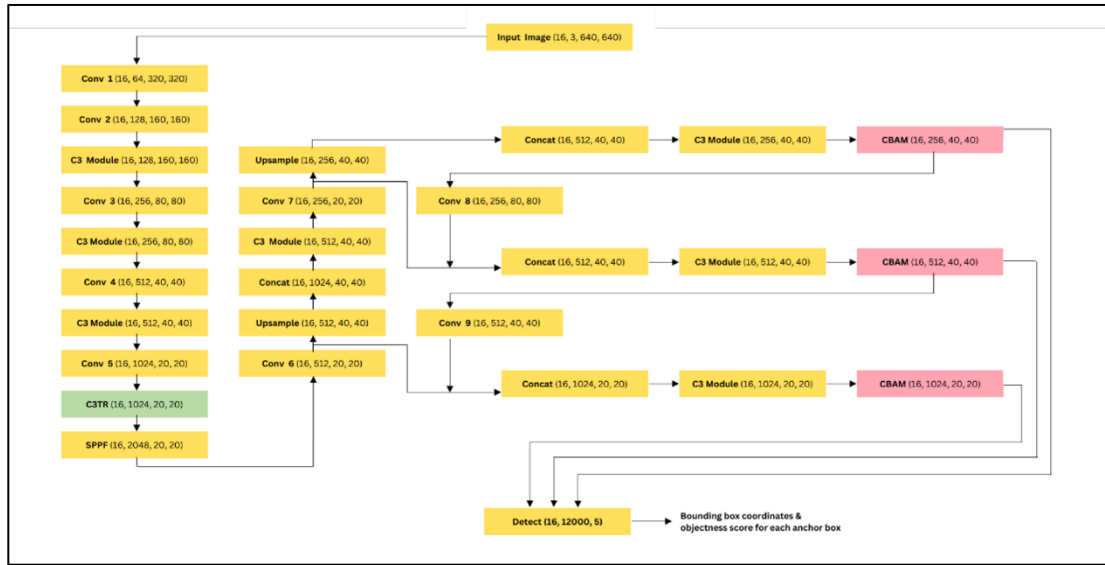


Figure 3: Improved YOLOv5x architecture diagram

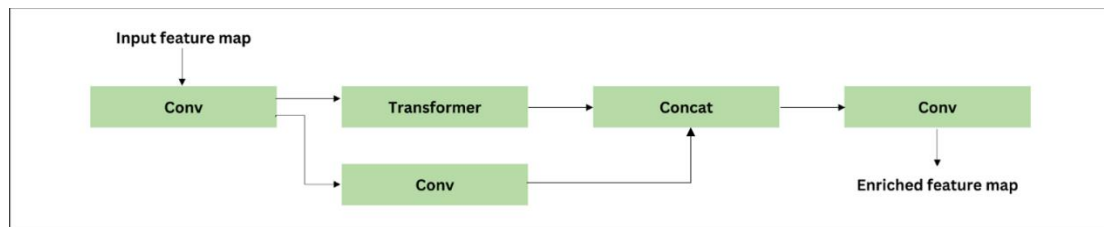


Figure 4: Architecture of C3TR module

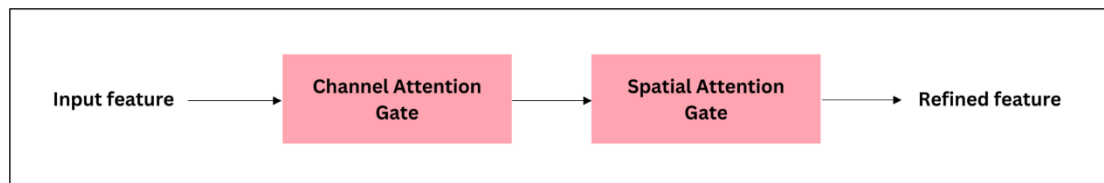


Figure 5: Architecture of CBAM module

An improved version of YOLOv5 model used in our project for the small object detection task. The architecture diagram of the same is given in Figure 3. The base model YOLOv5, was released under one of five size variants based on the number of network layers and parameters n (nano), s (small), m (medium), l (large) and x (extra-large). In our work we have adapted the YOLOv5x model as it has the highest mAP score for COCO dataset by a decent margin. The YOLOv5 model can be interpreted

as three broad components- backbone, neck and model head. The backbone network is primarily a convolutional network named Darknet53 which was already pre-trained on over a million images from the ImageNet dataset. Cross Stage Partial (CSP) network strategy is applied on Darknet53, which helps solve the problem of redundant gradients by truncating the gradient flow to the deeper layers of the network [39]. This also reduces the number of network parameters involved hence increasing the inference speed. In YOLOv5, the neck architecture consists of a series of multi-scale feature fusion modules that merge the features from different levels of the backbone network. The first module combines the output from the lowest level of the backbone network with the output from the highest level, creating a high-resolution feature map with rich semantic information. This map is then passed through a series of additional feature fusion modules that gradually reduce the resolution of the features while increasing their semantic richness. The final output of the neck architecture is a set of feature maps with different scales, each of which contains object detection predictions at a different level of detail. These feature maps are then used by the detection head to generate the final object detection results. The head is responsible for generating the final object detection output, predicting the location, class, and confidence scores of objects in the input image. It consists of detection layers that predict bounding boxes using anchor boxes of different sizes, and also predict class probabilities and confidence scores for each detected object.

In addition to the base YOLOv5x architecture, for the purpose of our project, we introduce a C3TR module and CBAM module as highlighted in *Figure 3*. A C3TR module is a type of feature extraction module designed to extract high-quality features from the input image while reducing the computational cost [40]. The C3TR module, as showing in *Figure 4* works by first performing a series of convolutional operations on the input feature map to extract information at different scales. It then applies cross-scale channel and spatial attention mechanisms to selectively weight and combine the features across different scales. Finally, it uses a tiny shift operation to reduce the computational cost of the module. Convolutional Block Attention

Module (CBAM), is an attention mechanism that helps the network focus on relevant parts of the image and ignore the irrelevant parts [37]. The CBAM module, as shown in *Figure 5* consists of two main components: the channel attention module (CAM) and the spatial attention module (SAM). The CAM computes the channel-wise attention by learning the importance of each channel in the feature map, while the SAM computes the spatial attention by learning the importance of each spatial location in the feature map. In our YOLOv5 model, the CBAM module is used in the head architecture to help the network attend to relevant regions in the feature maps. The CAM and SAM modules are applied separately to the feature maps, and their results are then multiplied together to obtain the final attention map, which is used to weight the feature map before passing it to the next layer. This process helps the network focus on the most relevant information in the feature maps, leading to improved accuracy in object detection.

The starting coordinates of the bounding box denoted by x and y are calculated according to the *Equation 1* and *Equation 2* respectively. The width of the target bounding box is computed as per *Equation 3*. *Equation 4* is used to compute the height of target object bounding box.

$$b_x = (2 \cdot \sigma(t_x) - 0.5) + c_x \quad \text{Equation 1}$$

$$b_y = (2 \cdot \sigma(t_y) - 0.5) + c_y \quad \text{Equation 2}$$

$$b_w = p_w \cdot (2 \cdot \sigma(t_w))^2 \quad \text{Equation 3}$$

$$b_h = p_h \cdot (2 \cdot \sigma(t_h))^2 \quad \text{Equation 4}$$

Where b_x , b_y , b_w , and b_h , denote the bounding box coordinates, c_x , c_y is the cell offset from the top left corner along the x-axis and y-axis and, p_w and p_h are the prior width and height of the bounding box respectively. Considering the three outputs of YOLOv5 model namely the bounding boxes, the classes of the objects and the objectness scores (also known as confidence score), the loss function adopted for the model is given in *Equation 5*.

$$\text{Model loss} = \lambda_1 \cdot L_{\text{class}} + \lambda_2 \cdot L_{\text{objectness}} + \lambda_3 \cdot L_{\text{location}} \quad \text{Equation 5}$$

Where the total loss is computed as a sum of Binary Cross Entropy computed for classes loss and objectness loss, and Complete Intersection Over Union loss computed over the location loss.

CHAPTER – 5

SYSTEM IMPLEMENTATION

The project was implemented in Python 3.8.16 using Google Colaboratory. The following section is divided into the following sections: Section 5.1 describes the data preparation process for YOLO; Section 5.2 covers the model training and the parameters used; Section 5.3 describes the evaluation metric and finally, Section 5.4 gives an overview of the modular implementation of the project.

5.1 Data Preparation

The dataset used in this project was obtained from the ORCA-Uboat website, a Chinese USV-manufacturing company. The dataset, named FloW-IMG, contains a large collection of floating bottle images with PASCAL Visual Object Classes XML annotated labeling. The PASCAL VOC XML format aims to standardize the annotation format. *Figure 6* shows a sample of the XML annotations. The size tag provides information about the source image's resolution, and the depth tag is set to 3 because each image has 3 RGB channels. The name tag contains the class name, which is "bottles" in this case, as the dataset only contains this type of object. Therefore, the objective is to detect and localize the objects, rather than classify or segment them. The pose tag is unspecified since the aim is to localize objects where the orientation may vary highly, so the pose information will not provide much value. The truncated tag indicates the presence of the object, whether it is fully visible or partially visible. The bndbox tag contains all the necessary information related to the bounding box, including the starting and ending points in the X and Y axes, height, and width.

The YOLOv5 model was selected for small object detection. YOLO models have a standard input format known as YOLO annotation, which is a text file with numeric values representing the boundary box. However, the data obtained from the ORCA-

Uboat website only contains XML annotations. Therefore, the first step is to convert the annotation data into the format accepted by YOLO. This conversion can be easily accomplished using parsing. A simple Python code was written to read the XML file and parse it to generate the required text file with YOLO standard annotations.

To parse the XML annotations into YOLO format, it is necessary to understand what YOLO annotation is. The YOLO format represents each object in a new line or row in the text file. Each line contains the class of the object, the X and Y coordinates of the center point of the bounding box, and the height and width of the bounding box. All parameters related to the bounding boxes are normalized to fit within the range of 0 to 1.



```

<annotation>
  <folder>myPerson</folder>
  <filename>4266507720.jpg</filename>
  <size>
    <width>1280</width>
    <height>640</height>
    <depth>3</depth>
  </size>
  <object>
    <name>bottle</name>
    <pose>Unspecified</pose>
    <truncated>0</truncated>
    <difficult>0</difficult>
    <bndbox>
      <xmin>873</xmin>
      <ymin>551</ymin>
      <xmax>1142</xmax>
      <ymax>629</ymax>
    </bndbox>
  </object>
  <object>
    <name>bottle</name>
    <pose>Unspecified</pose>
    <truncated>0</truncated>
    <difficult>0</difficult>
    <bndbox>
      <xmin>31</xmin>
      <ymin>332</ymin>
      <xmax>105</xmax>
      <ymax>360</ymax>
    </bndbox>
  </object>
</annotation>

```

Figure 6: A sample from the FloW-IMG dataset with its XML annotations

5.2 Training

The first step in our implementation was to prepare the YOLO standard annotations for the dataset of floating bottles in inland water bodies. All small objects in this implementation were defined as objects covering at most 32 x 32 pixels of an image.

YOLOv5 model was chosen for its superior accuracy compared to its predecessors and because it had already been pre-trained on the COCO dataset. Transfer learning is utilized and the pre-trained model is fine-tuned on our dataset. The dataset consisted of 2000 images and annotations, which was split in the ratio of 3:2 for training and testing. As there was only one class of objects in the Flow dataset, which was bottle, the objective was to detect and localize them. The model was initialized with the pre-trained weights that were obtained by training on the COCO dataset. The architecture was designed to be suitable for training with an input image size of 640 x 640 pixels. The model's training parameters are listed in *Table 1*. To further improve the performance of our model, various data augmentation techniques, including HSV, translation, scaling, horizontal flipping, and Mosaic were utilized as mentioned in *Table 2*. The HSV augmentation was used with a range of values, including a hue shift, saturation shift, and value shift. The translation, scaling and horizontal flipping augmentations were applied randomly. Mosaic augmentation was used over full training data to increase the diversity of the training. The training process was carried out on Google Colab Pro, utilizing the GPU accelerator to expedite the training process. During training, the model was evaluated using the validation dataset to ensure that it was not overfitting to the training dataset.

Table 1: Hyperparameter used in model training

Hyperparameter	Value
Learning Rate	0.01
Momentum	0.937
Weight Decay	0.0005
Warmup Epoch	3
Warmup Momentum	0.8
Optimizer	Stochastic Gradient Descent
Epochs	100
Batch Size	16
Image Size	640
Dropout	0.3

Table 2: Augmentation used in model training

Augmentation	Value
HSV	0.015, 0.7, 0.4
Translate	0.1
Scale	0.5
Flip left-right	0.5
Mosaic	1.0

5.3 Metrics

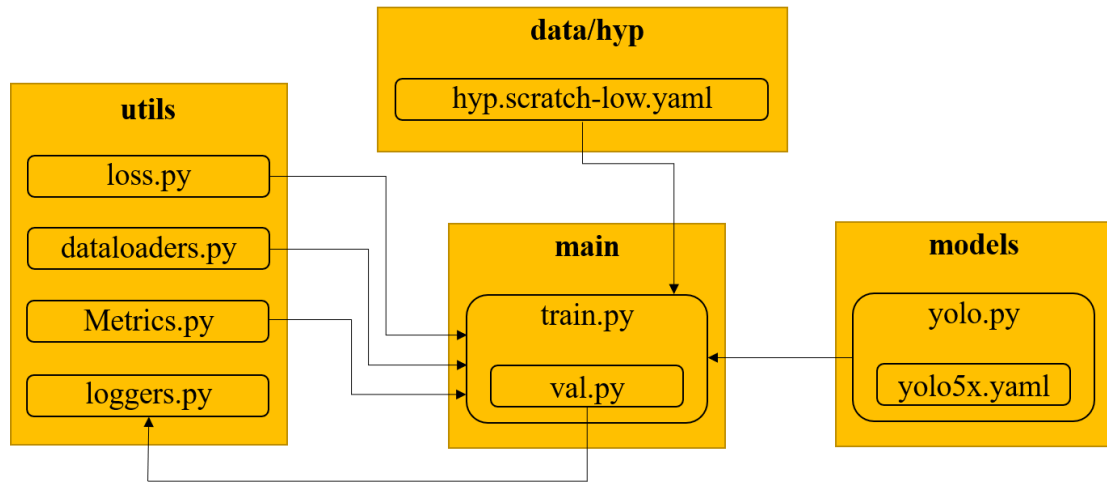
Metrics are an essential aspect of evaluating the performance of machine learning models. Metrics provide quantitative measures to assess the accuracy, precision, recall, and other aspects of a model's performance. One of the widely used metrics for object detection models is mean average precision (mAP). mAP is a measure of how well a model detects objects within an image, considering both precision and recall. The metric is calculated by averaging the precision values at various recall levels. A commonly used variation of mAP is the 0.5:0.95 mAP, which considers precision values over a range of IoU (Intersection over Union) thresholds, from 0.5 to 0.95. This metric is particularly useful for evaluating object detection models since it takes into account a range of object sizes, shapes, and orientations, providing a more comprehensive assessment of the model's performance.

5.4 Modules

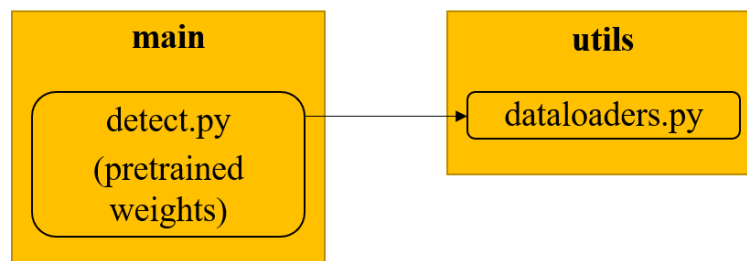
The proposed system is designed as a distributed architecture, with multiple modules communicating with each other to function as an ensemble, as illustrated in Figure 7.a. During the training phase, the main folder contains a Python file named "train.py" responsible for model training. To begin training, the "yolo.py" file in the "models" folder is used to load the YOLO model specified in the "yolov5x.yaml" file. The required training data is imported to the model using "DataLoader.py" in the "utils" folder, and training begins. The "train.py" file also calls "val.py" to validate

the model's learning after each iteration. The "metrics.py" file in "utils" calculates metrics such as mAP, mmAP, recall, precision, and losses, computed by "loss.py". These metrics are returned to "train.py" and logged using "loggers.py" in "utils", updating the Comet ML platform for visualization and generating a log file containing all necessary information, including the model's weights.

During the detection or inference phase, as shown in Figure 7.b, the previously saved weights from the training phase are loaded using "detect.py" in the main folder. Similar to the training phase, "DataLoader.py" is used to load the images for detection, and "detect.py" detects and localizes objects using the trained weights. The images and progress are stored as logs for later analysis.



7.a



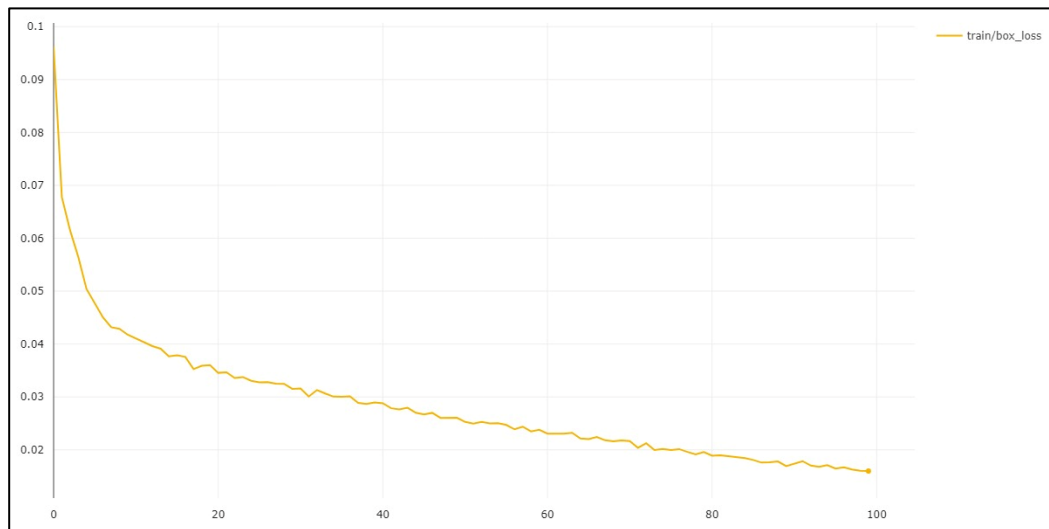
7.b

Figure 7: Modular workflow of code implementation

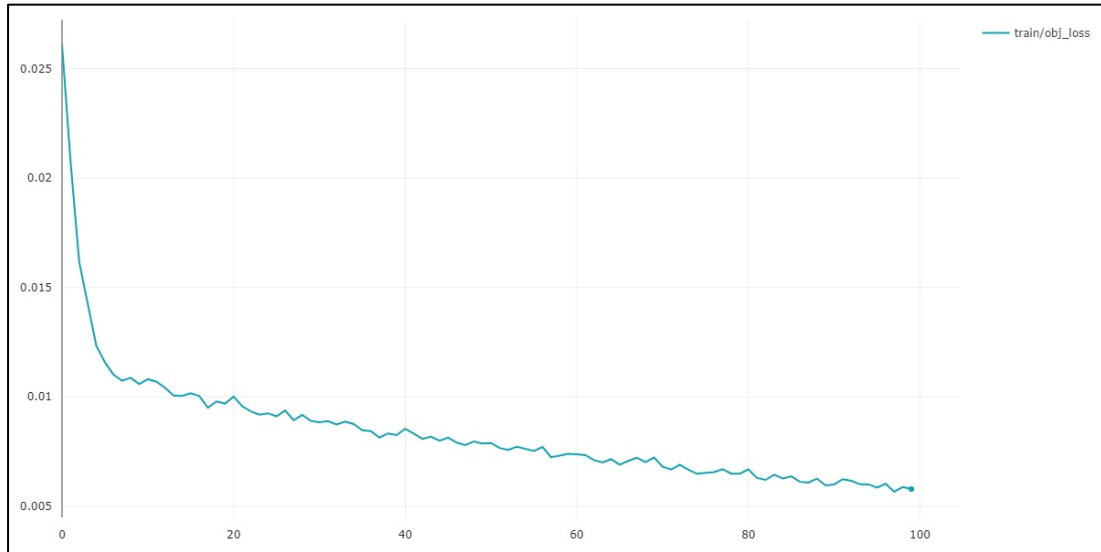
CHAPTER - 6

SYSTEM TESTING

The proposed system is tested on 800 images from the test set of the FloW-IMG dataset. YOLOv5 uses the summation of three losses as its loss function, namely box loss, location loss and objectness loss as described in Chapter 4. In our case the class loss will be 0 as we only have one object class. The plots of the training losses and validation losses are given in *Figure 8* and *Figure 9* respectively. It can be deduced from these plots that the training and validation losses have reduced as desired. We look at three metrics namely the mean average precision scores (mAP 0.5 and mAP 0.5:0.95), recall and precision. Graphs for the same are plotted and visualized using ClearML and Comet as given in Figure 8.

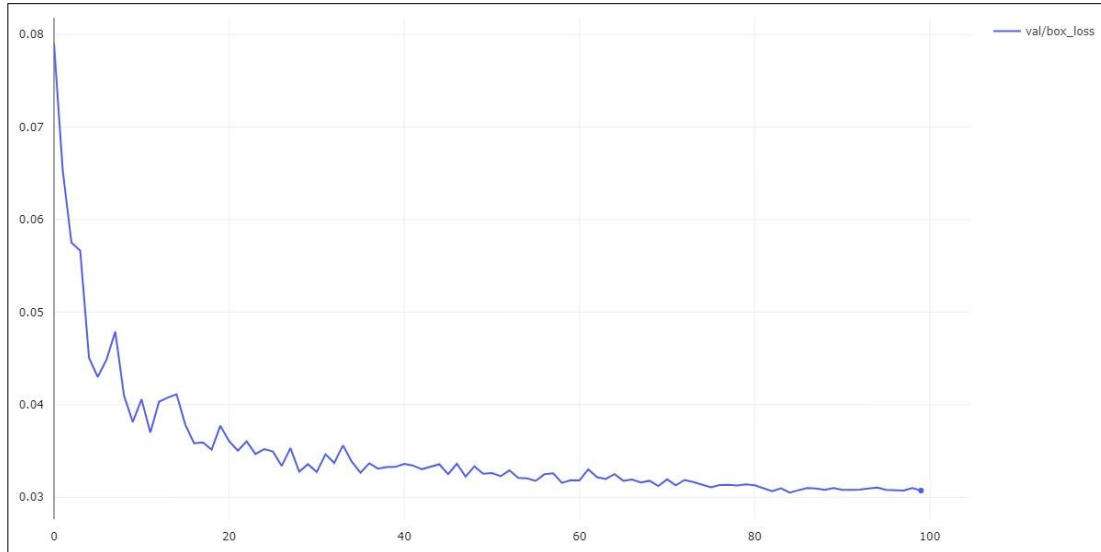


8.a

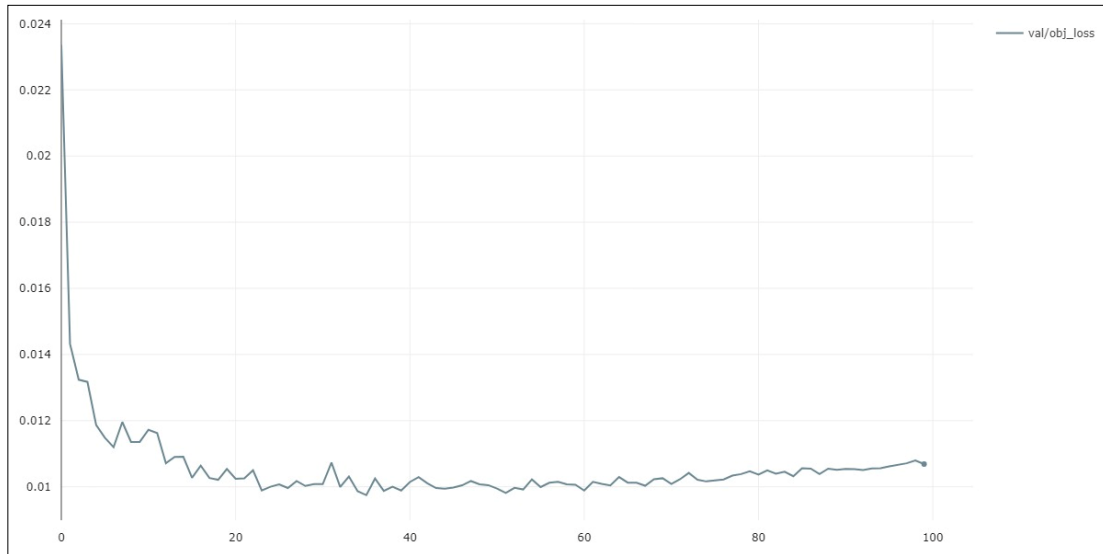


8.b

Figure 8: Training loss plots of the model. 8.a shows the training box loss and 8.b shows the training objectness loss.



9.a



9.b

Figure 9: Validation loss plots of the model. 9.a. shows the validation box loss and 9.b shows the validation object loss.

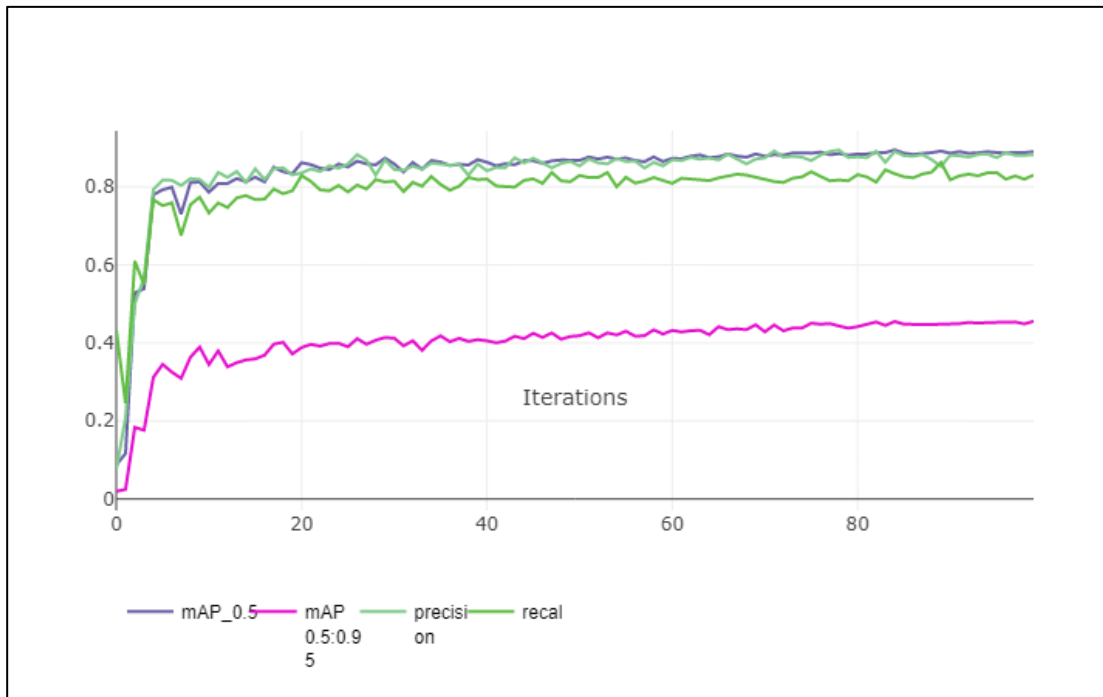


Figure 10: Plot containing test metrics- mAP 0.5, mAP 0.5:0.95, precision and recall

Further tests have been carried out by testing the proposed model with some images that have been captured from a lake in Bangalore to understand the model's ability to generalize. The performance on these images has been studied with respect to the specific challenges as discussed in Chapter 4.

CHAPTER – 7

RESULTS AND ANALYSIS

Our proposed model achieved a mAP score of 45.6% which is 1.8% increase in accuracy as compared to the baselines proposed in [3] for floating bottle detection. A comparison of proposed model's performance and presented baselines for the FloW-IMG dataset is given in *Table 3*.

Table 3: Proposed model performance comparison with models proposed in [3]

Model	mAP[0.5:0.95]
DSSD	0.275
RetinaNet	0.249
YOLOv3	0.128
Faster R-CNN	0.184
FPN	0.334
Cascade R-CNN	0.434
YOLOv5x	0.450
<i>Improved YOLOv5 (Proposed)</i>	<i>0.456</i>

The results section can be further divided and studied under the following sections: In Section 7.1 the model's prediction on the test dataset is analyzed; Section 7.2 studies the model predictions on the image data collected from nearby lake; and finally, Section 7.3 compares the effect of activation functions on the model's performance.

7.1 Analyzing model predictions on test set



11.a



11.b



11.c



11.d



11.e



11.f



11.g



11.h



11.i



11.j



11.k

Figure 11: Sample model predictions on images from the test dataset

The images presented in *Figure 11* are representative of the test set and similar to those in the training set. Images *11.a* and *11.e* depict high light reflections on the water surface, and the model demonstrates accurate predictions that are robust to surface reflections. However, in image *11.a*, the model's confidence score is relatively low due to additional factors such as the small size and lower clarity of the target object. Images *11.b*, *11.g*, *11.h*, and *11.k* exhibit multiple occurrences of bottles at different complexities. In images *11.b* and *11.h*, where all the bottles are distinctly visible and space separated, the model accurately identifies all. In image *11.g*, where

the bottles form a small cluster with some degree of overlap, the model successfully distinguishes five out of six bottles, but the confidence scores are somewhat reduced. In image *11.k*, although the bottles are distinctly placed, their size is significantly diminished, resulting in the correct identification of only a fraction of the number of bottles present. Images *11.c* and *11.f* demonstrate the model's performance against a camouflage scenario, where the model misses detecting the bottles. This is due to the bottles being well-hidden in the background and difficult to distinguish even with the naked eye. Image *11.d* depicts the correct identification of target bottles, despite their extremely small and diminished size. However, the model also makes some false predictions, which can be attributed to the lack of information in the image that helps the model distinguish bottles from other objects. Another example of false prediction is image *11.j*, where the model acquires insufficient context from the image due to occlusion. Images *11.d* and *11.i* are cases where the target object is over-lit. However, in image *11.i*, the bottle being much larger provides the model with a clearer context, resulting in accurate detection.

Based on the analysis of the model's performance on the images presented in *Figure 11*, it can be concluded that the model is robust to high light reflections on the water surface and can accurately detect bottles of different complexities when they are still distinctly visible. However, the model struggles in scenarios where the bottles are camouflaged or hidden in the background. Additionally, the model's performance is affected by occlusion, lack of information, and the size and clarity of the target object.

7.2 Testing on outside-dataset images



12.a



12.b



12.c



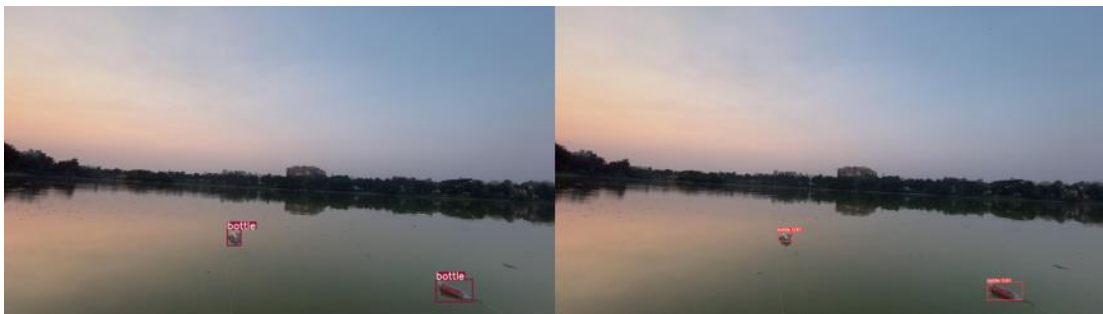
12.d



12.e



12.f



12.g



12.h



12.i



12.j

Figure 12: Sample model predictions on images that are collected from Bangalore lakes.

Consider the images presented in *Figure 12*. The model accurately identifies almost all occurrences of bottles on the water surface. Images *12.d* and *12.h* present examples of missed prediction. In image *12.d*, it can be noticed that there is an overlap of bottles where as in image *12.h* the shape of the bottle is distorted. However, it can be noticed that the model makes a lot of false predictions as shown in images *12.a*, *12.b*, *12.c*, *12.d* and *12.i*. In all these figures it can be noticed that the model seems to be confused with other objects of similar physical features.

7.3 Experimental analysis of Swish, Leaky ReLU and Hardswish activations on model performance

One important aspect of building a successful deep learning model is choosing the right activation function. In this work, we compare three activation functions: Swish, FReLU, and Hardswish, in the proposed model. Swish is a non-linear activation function that has been shown to outperform traditional activation functions such as ReLU in terms of accuracy. The equation of Swish activation is as given by *Equation 6*. Leaky ReLU is a variant of the ReLU activation function that avoids the "dying ReLU" problem by introducing a small slope for negative input values. The equation of Leaky ReLU activation is as given by *Equation 7*. Hardswish, on the other hand, is a computationally efficient non-linear activation function that has been recently proposed as an alternative to Swish. The equation of Hardswish activation is as given by *Equation 8*. Our experiments as represented in *Table 4*, show that the Hardswish activation function outperformed the other two activation functions in terms of accuracy and speed.

$$f(x) = x * (1 + e^{-x})^{-1} \text{ Equation 6}$$

$$f(x) = \begin{cases} 0.01x, & x < 0 \\ x, & x \geq 0 \end{cases} \text{ Equation 7}$$

$$f(x) = \begin{cases} 0, & x \leq -3 \\ x, & x \geq +3 \\ x \frac{x+3}{6}, & \text{otherwise} \end{cases} \text{ Equation 8}$$

Table 4: Impact of Different Activation Functions on Model Performance

Activation function	mAP[0.5:0.95]
Leaky ReLU	0.421
SiLU (Swish)	0.430
Hardswish	0.456

CHAPTER –8

CONCLUSION AND FUTURE SCOPE

The improved YOLOv5-based small object detector for detecting floating bottles in inland water bodies is able to predict bounding boxes for the test set of FloW-IMG dataset with a best mAP of 0.456. Three activation functions were compared, where Hardswish activation when paired with the proposed model gave the highest mAP. It can be established from the training and validation plots that the presented based on transfer learning has learned the FloW-IMG image data well. It can be inferred from the metrics plot that the model has a high recall rate signifying that the model successfully detects most of objects of interest. The precision value achieved by the model is also decently high which means the number of true positives is also higher. The model's generalization capability has also been tested on images captured from a nearby lake, where our model achieves a mAP of 0.316. Deducing from the presented results, it can also be stated that our proposed model performs well even in the presence of light reflections on the water surface. The model doesn't completely fall behind in case of diminished small objects, it manages to capture a few them correctly. The model is robust to blurred images and also manages to perform considerably well with over-sharpened and high contrast images. However, our model faces challenges when the small objects occur in a camouflaged or cluttered fashion within the given input image. Not all overlapped objects of interests get captured in the proposed model's predictions.

In the future there is significant scope for improvement in this project. The best accuracy achieved by our model is 45.6% which still has a huge scope of improvement. The model detection is not very robust to camouflage, this calls for future research and experiments. Model can be exposed to more images with cluttered and overlapping scenarios. Segmentation techniques can be looked upon for

the same. The application of the detection model can also be scaled by incorporating more classes of floating debris.

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